Junaid Muhammad

M.S. (Info. & Comm. Engg.) | Focused in AI and Computer Vision
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Jiho-ro, Jisan-dong, Dong-gu, Gwangju - 61445, South Korea

Objective

Master's graduate specializing in artificial intelligence for computer vision and medical image analysis, with hands-on experience in deep learning models such as CNNs, diffusion models, and transformer-based architectures, as well as clinically validated dataset curation. Research interests focus on AI-driven solutions for real-world problems across computer vision and healthcare applications.

Experience

- **Chosun University [🌐]** *Mar 2024 - Feb 2026*
  Graduate Research Assistant Gwangju, South Korea
  - Conducted research in medical image processing and image restoration.
  - Lead the curation of a clinically validated musculoskeletal multimodal dataset in collaboration with medical professionals, and developed a survival analysis framework for predicting fracture risk.
  - Developed Residual-CNN based deep learning frameworks for bone and hip fracture detection using pytorch.
  - Worked on transformer and diffusion based architecture for image processing.
- **University of Swat [🌐]** *Jan 2020 - Feb 2024*
  Administrative Assistant Swat, Pakistan
  - Academic and administrative tasks covering student admissions, records management, examination and results processing, transcript, degree and certificate issuance.
- **Spring Field School** *July 2018 - Dec 2019*
  Computer Operator Swat, Pakistan
  - Worked as IT In-charge, Teacher and Exam Invigilator, Managed IT Infrastructure of the school, Results and certificates processing, Fee Slips and Marks Certificates of students through ERP
- **NED University of Engineering and Technology [🌐]** *Jul 2017 - Jun 2018*
  Internee / Lab Engineer Karachi, Pakistan
  - Worked at the Department of Telecommunications Engineering, Faculty of Electrical and Computer Engineering. Responsibilities included students guidance through lab experiments and demonstrations, teacher assistant and exam invigilation.

Education

- **Chosun University** *Mar 2024 - Feb 2026*
  M.S. Information and Communication Engineering Gwangju, South Korea
  - GPA: 4.11/4.50; Percentage: 95.32; Thesis: **A Study on Hierarchical Pyramid Context Refinement with Supervised Contrastive Learning for Hip Fracture Classification from Radiographs**; Thesis word count: 12156
- **Iqra National University** *Sep 2012 - Feb 2017*
  B.S. Telecommunication Peshawar, Pakistan
  - GPA: 2.96/4.00; Percentage: 75.79; Thesis project: DTMF controlled IP based live broadcasting robot with night vision - Thesis and end-to-end hardware and system development

Conference Papers Presentations

[1.] Junaid Muhammad, Bumshik Lee. (2025). **"Dual Attention Transformer for Image Super Resolution",** Oral presentation, 01-03 May, 2025, Korean Institute of Smart Media and E-Commerce (KISM) Spring Academic Conference, Seoul, South Korea.

[2.] Junaid Muhammad, Bumshik Lee. (2025). **"Bone Abnormality Classification with Supervised Contrastive Learning on MURA",** Oral presentation, 06-09 November, 2025, The 26th International Symposium on Advanced Intelligent Systems, Cheongju, South Korea.

Publications

C=Conference, J=Journal, P=Patent, R=In Revision, T=Thesis

[R.1] Junaid Muhammad, Bumshik Lee. (2026). **HPCRNet: A Hierarchical Pyramid Context Refinement Framework for Interpretable Hip Fracture Analysis**. *Applied Soft Computing (ASOC)*, In revision, 2026.

Skills

- **Programming Languages:** Python, C++
- **Data Science & Machine Learning:** PyTorch, TensorFlow, Huggingface, timm, OpenCV, NumPy, Scikit-learn, CUDA, Jupyter Notebook, PyCharm
- **Specialized Area:** Computer Vision, Image Processing, Deep Learning, Medical Image Analysis, Medical Imaging Dataset Curation, Transformers, Diffusion Models, Convolutional Neural Networks, Residual Learning
- **Research Skills:** End-to-End Research Workflow, Scientific Writing, Benchmarking, Strong Written & Communication Skills, Data Visualization

Honors and Awards

- **Chosun Unviersity Foreign Excellence Scholarship** *Mar 2024 - Feb 2026*
  Chosun University, Gwangju, South Korea

Certifications

- **IELTS Academic:** Overall Score 7.0 *Mar 2026*

Additional Information

Languages: English (Fluent), Urdu (Native), Pashto (Native), Korean (Basic Proficiency)
Interests: Hiking, Sports, Reading

References

1. **Dr. Arif Ullah**
   Research Fellow, Queen's University, Belfast, UK
   Email: a.ullah@qub.ac.uk
   GitHub
   Relationship: Former Assistant Professor, Dept. of Inormation and Communication Engineering, Chosun University

2. **Dr. Fawad**
   Postdoctoral Researcher, Korea Institute of Energy Technology (KENTECH), South Korea
   Email: Fawad1@kentech.ac.kr
   Google Scholar
   Relationship: Colleague at ENVI Lab, KENTECH, South Korea - Former Postdoctoral researcher at Chosun University